

Question No:8

DS	Session	Searching	Question Information	Level 1 Challenge 8
----	---------	-----------	----------------------	-----------------------

Problem Description:
Ram's first job is a out of university is for a student's book publisher. They are currently publishing a line of "intelligent libs" books, and they need you to check their print-copy. Write a program that will take a sentence with placeholders in it, and substitute in for those placeholders words from the lists of example words the publisher has provided.

Your goal will be to substitute the appropriate words into the sentence from the list of provided words TWICE (keeping track of the words you have used) where:

- [N] = nouns
- [AV] = adverbs
- [V] = verbs
- [A] = adjectives

Input Format:
Line one will always be the line with the story. The order of the Nouns, Adverbs, Verbs and Adjectives will always be the same. Input stops when it reaches the word END alone on a line. You are guaranteed to always have enough words to substitute into the sentence TWICE (not repeating any words). The number of words given under each section will be random.

Output Format:
Print the output in a separate lines contains, Always take the TOP-most UNUSED word from a group when substituting. Output the final sentences when all substitutions are done.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
squirrel
London
tree
lettuce
ADVERBS
silently
quickly

Test Case 2

INPUT (STDIN)

There once was a [AJ] [N] from [N] who [AV] [V] all day.
NOUNS
singer
chennai
hotel
chettinad
ADVERBS
silently
quickly

happily
VERBS
skipped
ran
climbs
ADJECTIVES
delightful
homely
quain

EXPECTED OUTPUT

There once was a delightful squirrel from London who
silently skipped all day.
There once was a homely tree from lettuce who quickly ran
all day.

happily
sadly
VERBS
skipped
reads
ran
sings
climbs
ADJECTIVES
delightful
homely
surprisingly

EXPECTED OUTPUT

There once was a delightful singer from chennai who silently
skipped all day.
There once was a homely hotel from chettinad who quickly
reads all day.

▼ Mandatory Test Cases

Test Case 1

KEYWORD

int cl[CMDS]

Test Case 2

KEYWORD

char *lists[CMDS][MAXWORDS];

Test Case 3

KEYWORD

char *tokens[TOKENS]={" [N]"," [AV]"," [V]"," [AJ]"};

Test Case 4

KEYWORD

char *cmds[CMDS]=
{ "NOUNS", "ADVERBS", "VERBS", "ADJECTIVE
S", "END"};

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

3

Test Case 2

TOKEN COUNT

488

Test Case 3

NLOC

50

Code:

```
1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s[1000];
6      char line[1000];
7
8      char v[4][100][100];
9      int count[4] = {0, 0, 0, 0};
10
11     fgets(s, sizeof(s), stdin);
12     s[strcspn(s, "\n")] = '\0';
13
14     while (fgets(line, sizeof(line), stdin)) {
15         line[strcspn(line, "\n")] = '\0';
16
17         if (strcmp(line, "END") == 0)
18             break;
19
20         if (strcmp(line, "NOUNS") == 0) {
21             while (fgets(line, sizeof(line), stdin)) {
22                 line[strcspn(line, "\n")] = '\0';
23
24                 if (strcmp(line, "ADVERBS") == 0)
25                     break;
26
27                 strcpy(v[0][count[0]++], line);
28             }
29         }
30     }
```

```
29     }
30
31     if (strcmp(line, "ADVERBS") == 0) {
32         while (fgets(line, sizeof(line), stdin)) {
33             line[strcspn(line, "\n")] = '\0';
34
35             if (strcmp(line, "VERBS") == 0)
36                 break;
37
38             strcpy(v[1][count[1]++], line);
39         }
40     }
41
42     if (strcmp(line, "VERBS") == 0) {
43         while (fgets(line, sizeof(line), stdin)) {
44             line[strcspn(line, "\n")] = '\0';
45
46             if (strcmp(line, "ADJECTIVES") == 0)
47                 break;
48
49             strcpy(v[2][count[2]++], line);
50         }
51     }
52
53     if (strcmp(line, "ADJECTIVES") == 0) {
54         while (fgets(line, sizeof(line), stdin)) {
```

```

55         line[strcspn(line, "\n")] = '\0';
56
57         if (strcmp(line, "END") == 0)
58             break;
59
60         strcpy(v[3][count[3]++], line);
61     }
62 }
63
64
65 char *key[4] = {"[N]", "[AV]", "[V]", "[AJ]"};
66 int p[4] = {0, 0, 0, 0};
67
68 for (int k = 0; k < 2; k++) {
69
70     char t[1000];
71     strcpy(t, s);
72
73     for (int i = 0; i < 4; i++) {
74
75         char *pos;
76
77         while ((pos = strstr(t, key[i])) != NULL) {
78
79             int index = pos - t;
80             int keyLen = strlen(key[i]);
81             int wordLen = strlen(v[i][p[i]]);

```

```
82
83         /* Move the remaining characters */
84         memmove(
85             t + index + wordLen,
86             t + index + keyLen,
87             strlen(t + index + keyLen) + 1
88         );
89
90         /* Insert the replacement word */
91         memcpy(
92             t + index,
93             v[i][p[i]],
94             wordLen
95         );
96
97         p[i]++;
98     }
99 }
100
101     printf("%s\n", t);
102 }
103     return 0;
104 }
```

Output:

Test Case-1:

There once was a [AJ] [N] from [N] who [AV] [V] all day.

NOUNS

squirrel

London

tree

lettuce

ADVERBS

silently

quickly

happily

VERBS

skipped

ran

climbs

ADJECTIVES

delightful

homely

quaint

There once was a delightful squirrel from London who silently skipped all day.

There once was a homely tree from lettuce who quickly ran all day.

Test Case-2:

	There once was a [AJ] [N] from [N] who [AV] [V] all day.	
	NOUNS	
	singer	
	chennai	
	hotel	
	chettinad	
	ADVERBS	
	silently	
	quickly	
	happily	
	sadly	
	VERBS	
	skipped	
	reads	
	ran	
	sings	
	climbs	
	ADJECTIVES	
	delightful	
	homely	
	surprisingly	
	There once was a delightful singer from chennai who silently skipped all day.	
	There once was a homely hotel from chettinad who quickly reads <u>all</u> day.	

Keshav Kumar Rauniyar
CH.SC.U4CSE25236